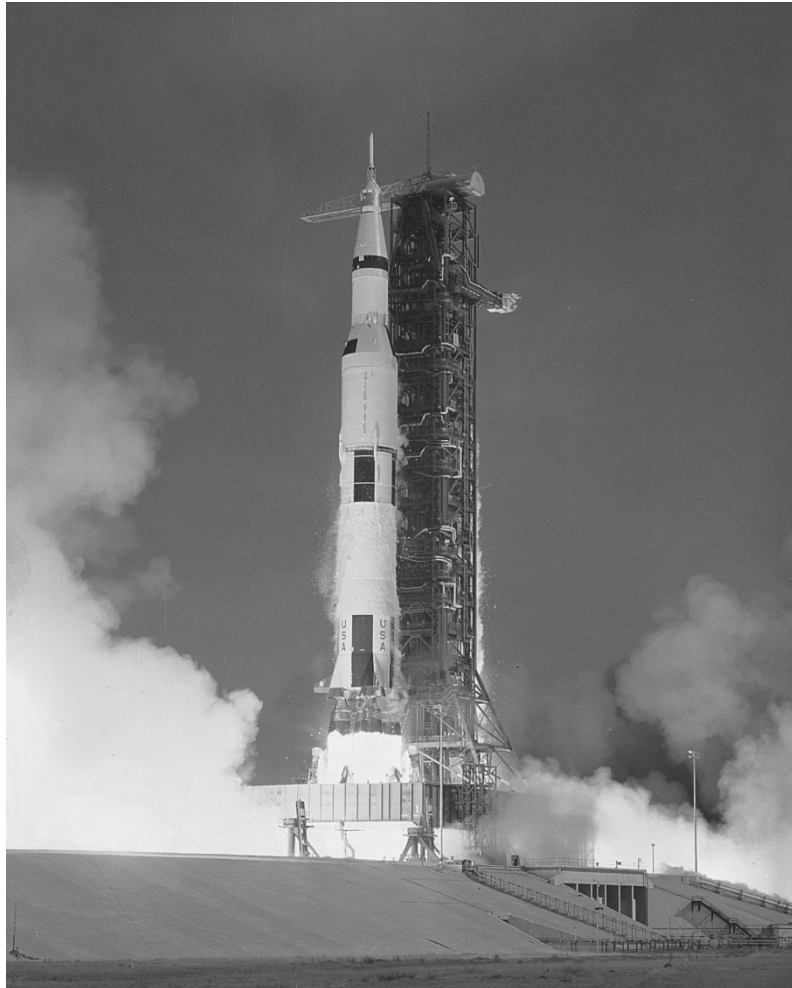


Chapter 3 Integral Relations for a Control Volume



Learning Objectives

- Recognize basic laws of fluid mechanics.
- Understand the Reynolds transport theorem and its role in deriving the basic control volume relations of fluid mechanics.
- Apply conservation of mass relation for fluid flows.
- Calculate velocity, pressure, and force for fluid flows using the linear momentum relation.
- Understand Bernoulli equation and its limitation in calculating momentum and energy.
- Apply the energy conservation equation to a fluid system including pump and turbine.

-
- **Fundamental Physics Laws**
 - **Control Volume, Volume Flow Rate, and Mass Flow Rate**
 - **Reynolds Transport Theorem**
 - **Conservation of Mass**
 - **Conservation of Linear Momentum Equation**
 - **Examples of Force Calculation Using Control Volume**
 - **Bernoulli's Equation**
 - **Conservation of Angular Momentum**
 - **Conservation of Energy Equation**

Basic Physical Laws of Fluid Mechanics

- Three basic approaches to the analysis of fluid mechanics problems:
 1. **Integral, or large-scale analysis in a finite control volume (Chap. 3).**
 2. **Differential, or small-scale analysis in an infinitesimal control volume (Chap. 4).**
 3. **Experimental, or dimensional analysis (Chap. 5).**
- The three approaches are equally important.

Basic Laws of Mechanics for a System

- All the laws of solid mechanics are written for a *system*, defined as an arbitrary quantity of mass of fixed identity.

- **Conservation of mass,**

$$m_{\text{sys}} = \text{const} \quad \text{or} \quad \frac{dm}{dt} = 0$$

- The Newton's second law (**linear momentum** equation),

$$\mathbf{F} = m\mathbf{a} = m \frac{d\mathbf{V}}{dt} = \frac{d(m\mathbf{V})}{dt}$$

- Moment and **angular momentum** relation,

$$\mathbf{M} = \frac{d\mathbf{H}}{dt} \quad \text{where the angular momentum } \mathbf{H} = \sum (\mathbf{r} \times \mathbf{V}) \delta m$$

- **Conservation of energy** (1st law of thermodynamics)

$$\delta Q - \delta W = dE \quad \text{or} \quad \dot{Q} - \dot{W} = \frac{dE}{dt}$$

Basic Laws of Mechanics for a System

- The second law of thermal dynamics,

$$dS \geq \frac{\delta Q}{T} \quad \left(\begin{array}{l} \text{The total entropy of an isolated system can never decrease} \\ \text{over time} \end{array} \right)$$

- State relations: **state relation** (or equation of state) expresses one thermodynamic property as a function of two other independent properties

$$p = p(\rho, T).$$

Ideal gas law:

Ideal gas is assumed to

- Have negligible volume
 - Have no intermolecular forces
 - Undergo perfectly elastic collision
- Energy-temperature relations with assumption of a calorically perfect gas

$$u = c_v T \quad \text{and} \quad h = c_p T$$

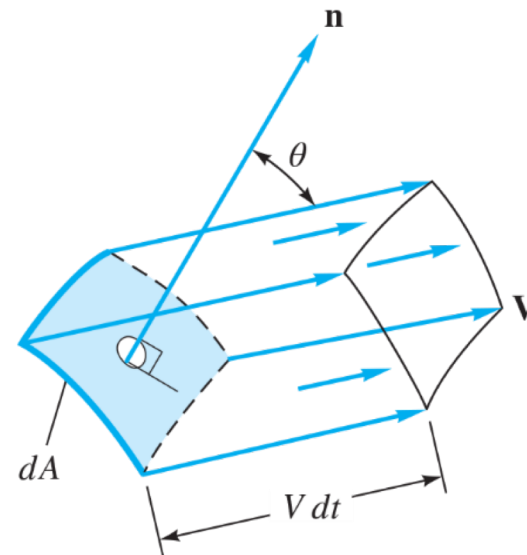
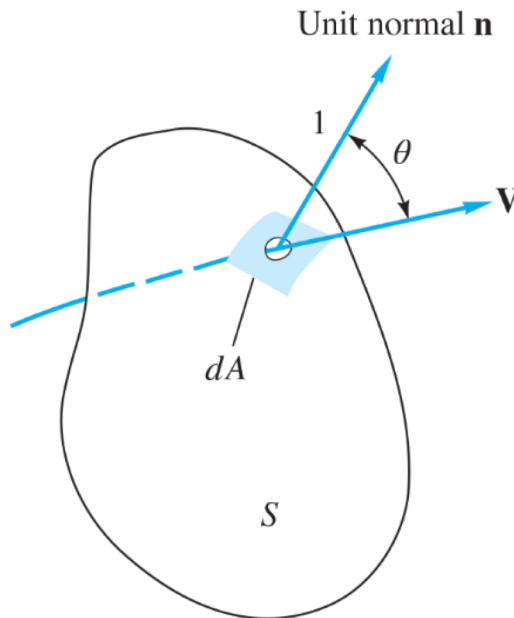
$$\gamma = \frac{c_p}{c_v}$$

Volume and Mass Rate of Flow

- Volume rate of flow (volume flow rate) is defined as the amount of fluid swept through dA in time dt (m^3/s),

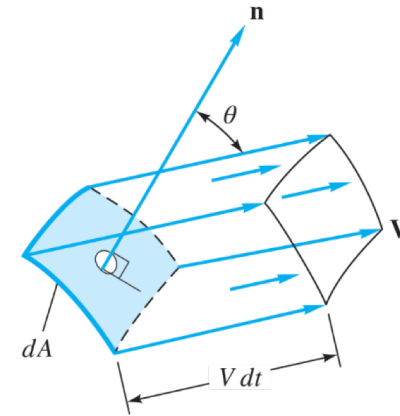
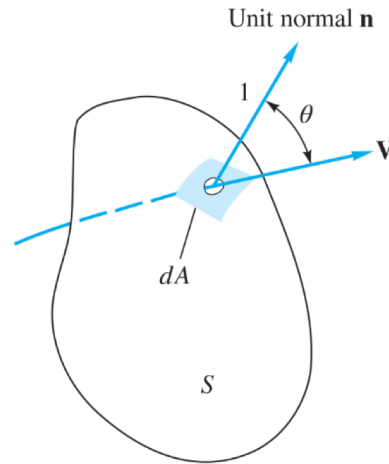
$$Q = \int_S \mathbf{v} \cdot \mathbf{n} \, dA$$

where $\mathbf{n}$ is unit vector normal to dA , positive when it is outward normal unit vector.



Volume and Mass Rate of Flow

$$Q = \int (\mathbf{V} \cdot \mathbf{n}) dA = \int V_n dA$$



- Volume flow can be multiplied by density to obtain

$$\dot{m} = \int \rho (\mathbf{V} \cdot \mathbf{n}) dA = \int \rho V_n dA$$

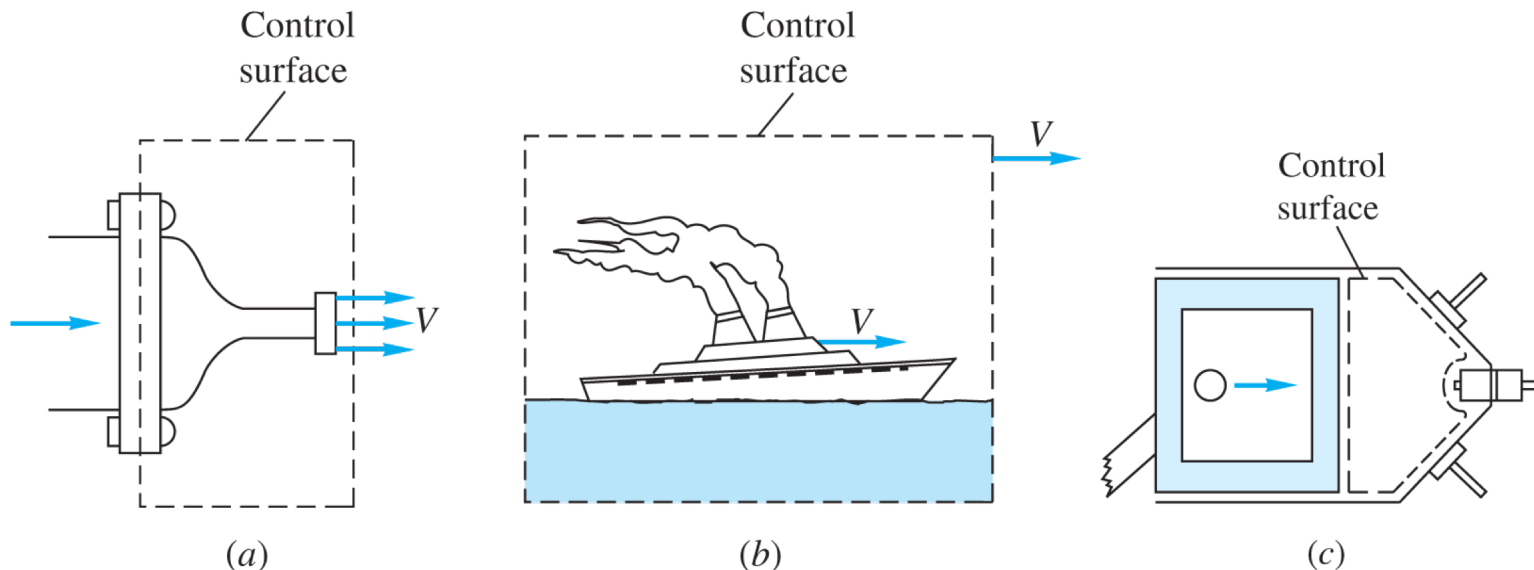
- If density and velocity are constant over the surface S ,

$$\dot{m} = \rho Q = \text{$$

- Units for Q are m^3/s , and for $\dot{m}$ are kg/s .

The Reynolds Transport Theorem

- *Reynolds transport theorem* that mathematically connects the **perspective** (tracking a specific, moving *system*) and the **perspective** (observing a fixed, stationary region in a *control volume*).
- The desired conversion formula differs slightly according to whether the control volume is fixed, moving, or deformable.



-
- System at time $t + dt$
- System at time t
- Fixed control volume CV
- dA
- θ
- $\mathbf{V}_{out}$
- $\mathbf{n}$, Unit outward normal to dA
- $\mathbf{V}_{in}$
- $\mathbf{n}$, Unit outward normal to dA
- Arbitrary fixed control surface CS
- $d\mathcal{V}_{in} = V_{in} dA_{in} \cos \theta_{in} dt = -\mathbf{V} \cdot \mathbf{n} dA dt$
- $d\mathcal{V}_{out} = V_{out} dA_{out} \cos \theta_{out} dt = \mathbf{V} \cdot \mathbf{n} dA dt$

Fixed, Moving and Deformable Control Volume

- Let B be any property of the fluid (mass, momentum, energy, etc.), and β is the amount of B per unit mass.

$$B_{CV} = \int_{CV} \beta dm = \int_{CV} \beta \rho dV \quad \text{and} \quad \beta = \frac{dB}{dm}$$

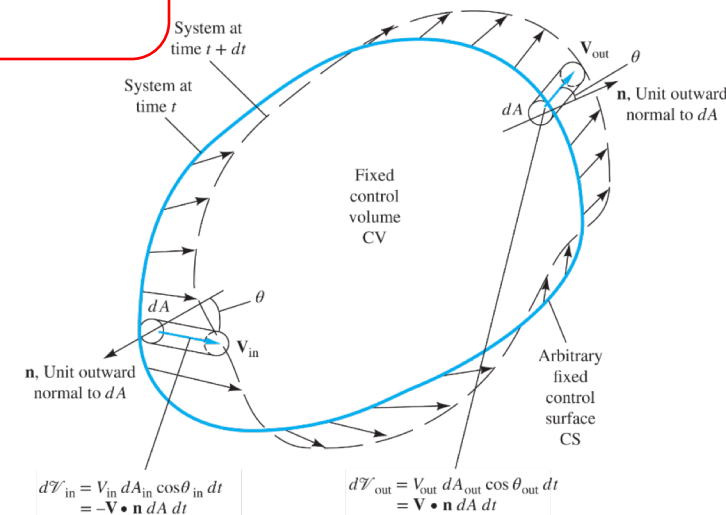
- The compact form of the Reynolds transport theorem is

$$\frac{dB_{sys}}{dt} = \frac{d}{dt} \left(\int_{CV} \beta \rho dV \right) + \int_{CS} \beta \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

- The flow terms of β from the CV are

$$\int_{CS} \beta \rho V \cos \theta dA_{out} - \int_{CS} \beta \rho V \cos \theta dA_{in}$$

Outflow of β from the CV
Inflow of β into the CV



Fixed, Moving and Deformable Control Volume

- Reynolds transport theorem for an arbitrary fixed control volume

$$\frac{dB_{\text{sys}}}{dt} = \frac{d}{dt} \left(\int_{\text{CV}} \beta \rho dV \right) + \int_{\text{CS}} \beta \rho V \cos \theta dA_{\text{out}} - \int_{\text{CS}} \beta \rho V \cos \theta dA_{\text{in}}$$

Outflow

Inflow

- Moving or deformable CV at $\mathbf{V}_s$,

$$\frac{dB_{\text{sys}}}{dt} = \frac{d}{dt} \left(\int_{\text{CV}} \beta \rho dV \right) + \int_{\text{CS}} \beta \rho \left(\boxed{} \right) dA$$

where relative velocity $\mathbf{V}_r$ is $\mathbf{V}_r = \mathbf{V} - \mathbf{V}_s$

For a fixed control volume, the volume elements do not vary, then

$$\frac{d}{dt} \left(\int_{\text{CV}} \beta \rho dV \right) = \int_{\text{CV}} \frac{\partial}{\partial t} (\beta \rho) dV$$

Conservation of Mass

- **Conservation of mass:**

For the Reynolds transport theorem, $B = m, \beta = 1$, for a fixed CV

$$\frac{dB_{\text{sys}}}{dt} = \frac{d}{dt} \left(\int_{\text{CV}} \beta \rho \, dV \right) + \int_{\text{CS}} \beta \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

$$\frac{dB_{\text{sys}}}{dt} = 0 = \frac{d}{dt} \left(\int_{\text{CV}} \rho \, dV \right) + \int_{\text{CS}} \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

$$\int_{\text{CV}} \frac{\partial \rho}{\partial t} dV + \int_{\text{CS}} \rho (\mathbf{V} \cdot \mathbf{n}) dA = 0$$

Conservation of Mass

- One-dimensional inlets and outlets

$$\int_{\text{CV}} \frac{\partial \rho}{\partial t} dV + \sum_i (\rho_i A_i V_i)_{\text{out}} - \sum_i (\rho_i A_i V_i)_{\text{in}} = 0$$

- One-dimensional and Steady flow,

$$\sum_i (\rho_i A_i V_i)_{\text{in}} = \sum_i (\rho_i A_i V_i)_{\text{out}}$$

Inflow Outflow

- One-dimensional, steady with one inlet and one outlet,

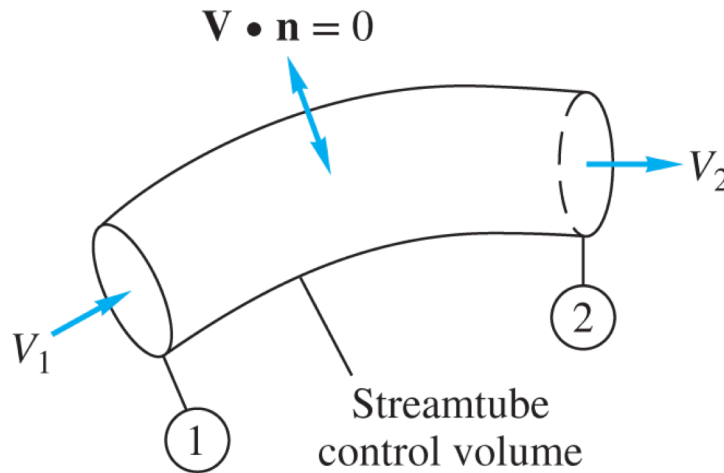
or $\dot{m}_1 = \dot{m}_2$

- One-dimensional, steady, Incompressible with one inlet/outlet $\rho = \text{const}$,

or $Q_1 = Q_2$

Conservation of Mass

Example: From the figure find exit velocity V_2 for steady flow through a stream tube, A_1 , A_2 , and V_1 are given.



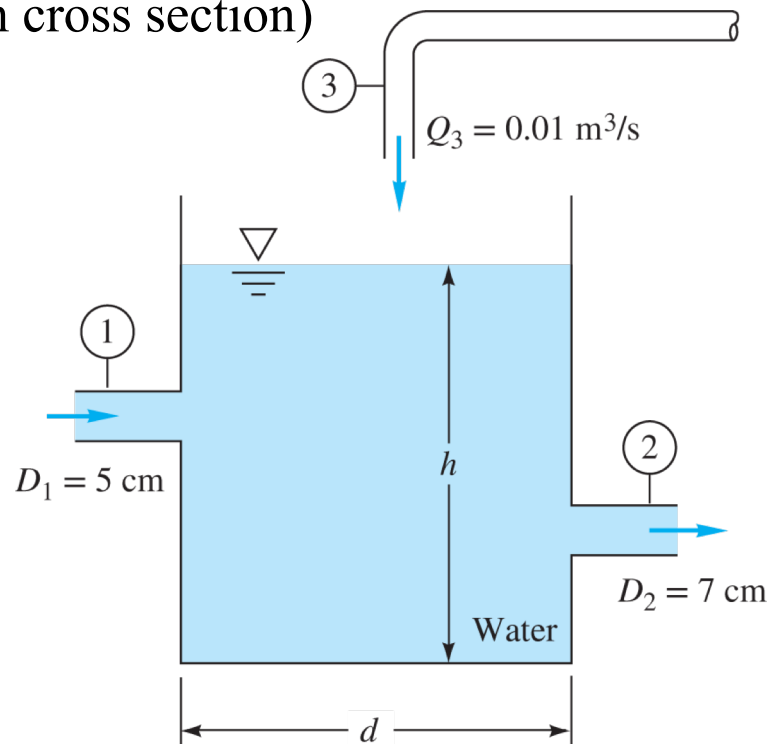
Solution: From conservation of mass,



Conservation of Mass

Example: The open tank contains water at 20°C and is being filled through sections 1 and 3. (1) Derive an expression for water level change dh/dt in terms of Q_1, Q_2, Q_3 , and d ; (2) for constant water level h , determine V_2 if $V_1 = 3$ m/s and $Q_3 = 0.01$ m³/s.

(Assume the tank is circular in cross section)



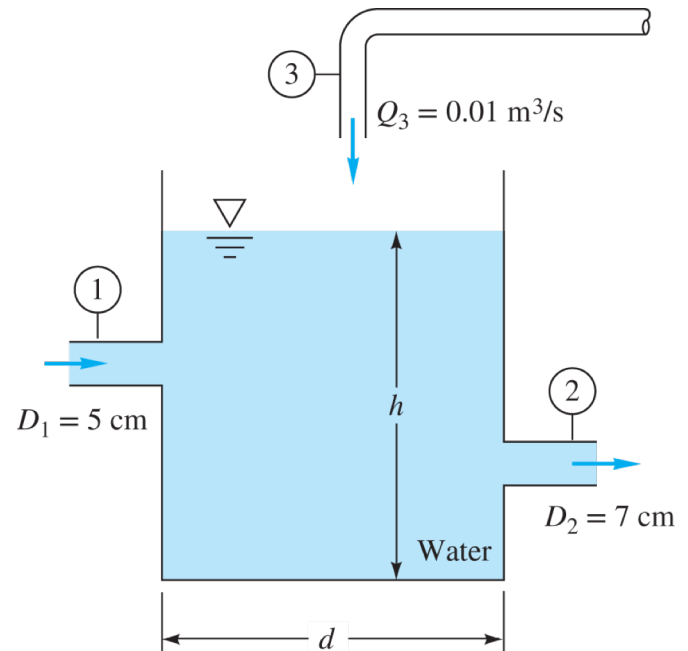
Conservation of Mass

Solution:

Select a CV enclosing water in the tank

Applying the mass conservation eq.

$$\frac{d}{dt} \left(\int_{CV} \rho dV \right) + \sum_i (\rho_i A_i V_i)_{out} - \sum_i (\rho_i A_i V_i)_{in}$$



-
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The Linear Momentum Equation

- The linear momentum: $\mathbf{B} = m\mathbf{V}$, $\beta = d\mathbf{B}/dm = \mathbf{V}$, fixed CV

$$\left(\frac{dB_{\text{sys}}}{dt} = \frac{d}{dt} \left(\int_{\text{CV}} \beta \rho dV \right) + \int_{\text{CS}} \beta \rho (\mathbf{V} \cdot \mathbf{n}) dA \right)$$

$$\frac{d(m\mathbf{V})_{\text{sys}}}{dt} = \sum \mathbf{F} = \frac{d}{dt} \left(\int_{\text{CV}} \mathbf{V} \rho dV \right) + \int_{\text{CS}} \mathbf{V} \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

$$\text{or } \sum \mathbf{F} = \frac{d}{dt} \left(\int_{\text{CV}} \mathbf{V} \rho dV \right) + \int_{\text{CS}} \mathbf{V} \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

- Force term $\sum \mathbf{F}$ should include all the surface forces (pressure, shear stress), volumetric body forces (gravity, buoyancy), and external forces.

The Linear Momentum Equation

- 1-D Momentum flux at inlets and outlets

$$\sum \mathbf{F} = \frac{d}{dt} \left(\int_{\text{CV}} \mathbf{V} \rho dV \right) + \sum_i \boxed{\phantom{\dot{m}_i \mathbf{V}_i}} - \sum_i \boxed{\phantom{\dot{m}_i \mathbf{V}_i}}$$

Momentum flow

- Steady and nondeforming flow,

$$\sum \mathbf{F} = \sum_i (\dot{m}_i \mathbf{V}_i)_{\text{out}} - \sum_i (\dot{m}_i \mathbf{V}_i)_{\text{in}}$$

- One inlet/outlet (1-inlet, 1-outlet, $\dot{m}_1 = \dot{m}_2 = \dot{m}$)

$$\sum \mathbf{F} = \boxed{\phantom{\dot{m}(\mathbf{V}_2 - \mathbf{V}_1)}}$$

Pressure Condition at a Jet Exit

- External pressure force on a closed control surface (CS) is normal to the surface and for the two cases shown

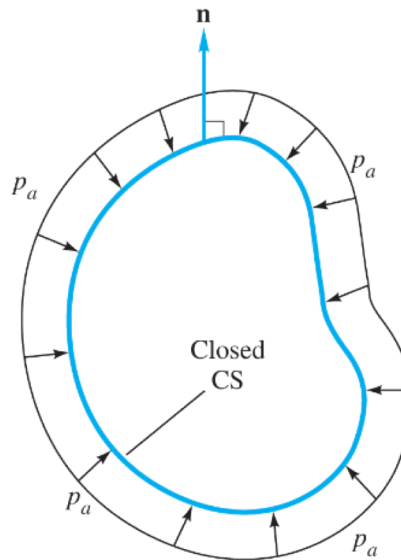
(a) uniform pressure

$$\mathbf{F} = -p_a \int \mathbf{n} dA \equiv 0$$

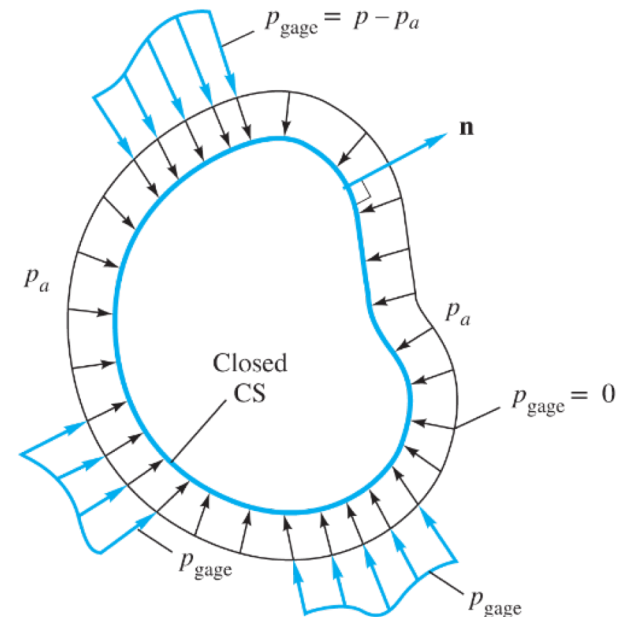
(b) nonuniform pressure

$$\mathbf{F} = \int (p - p_a)(-\mathbf{n}) dA \equiv 0$$

➔
$$\mathbf{F} = \int p_{gage}(-\mathbf{n}) dA \equiv 0$$



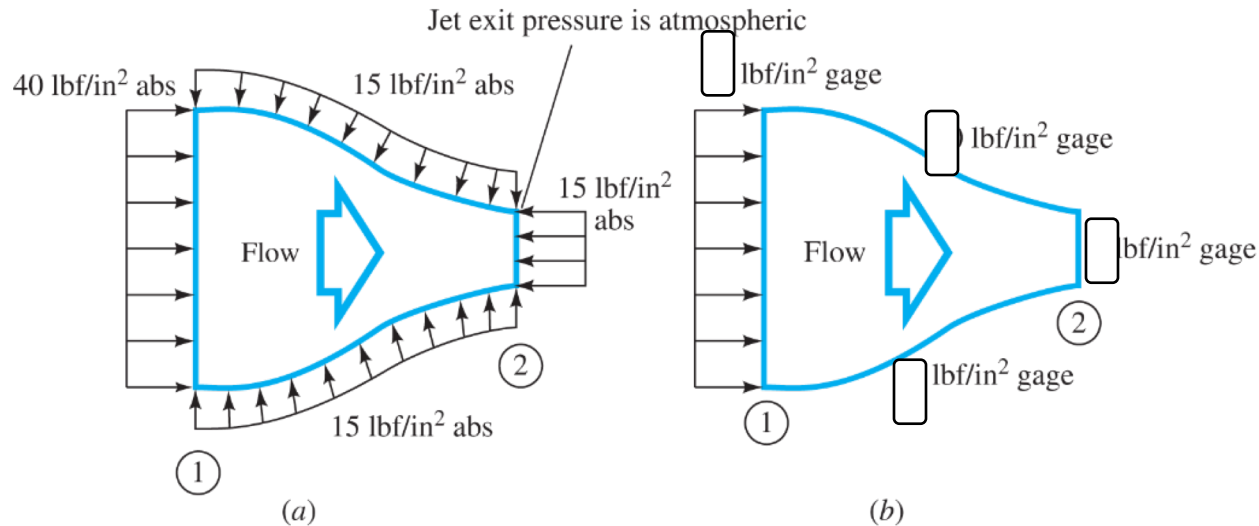
(a)



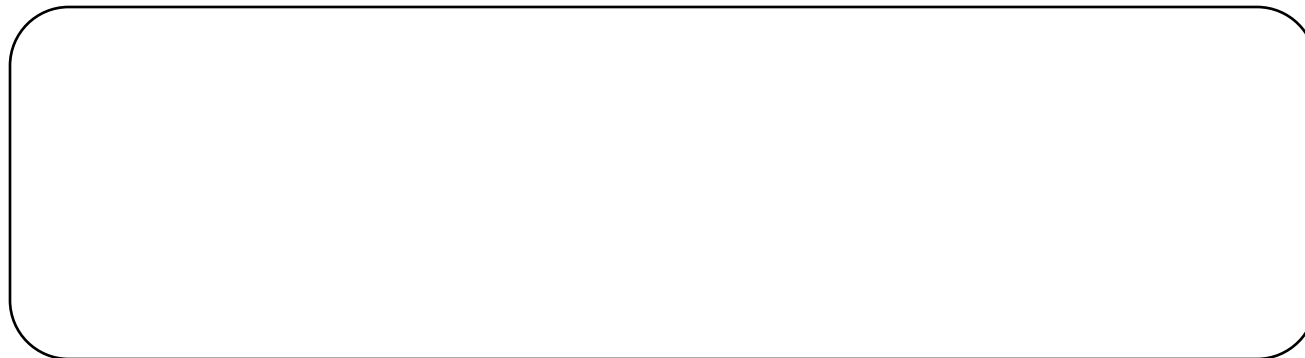
(b)

Pressure Condition at a Jet Exit

Example: For a nozzle section shown, find the net pressure if $D_1 = 3$ in.



Solution: Select the outside of the nozzle plus the cut sections (1) and (2) as CV,

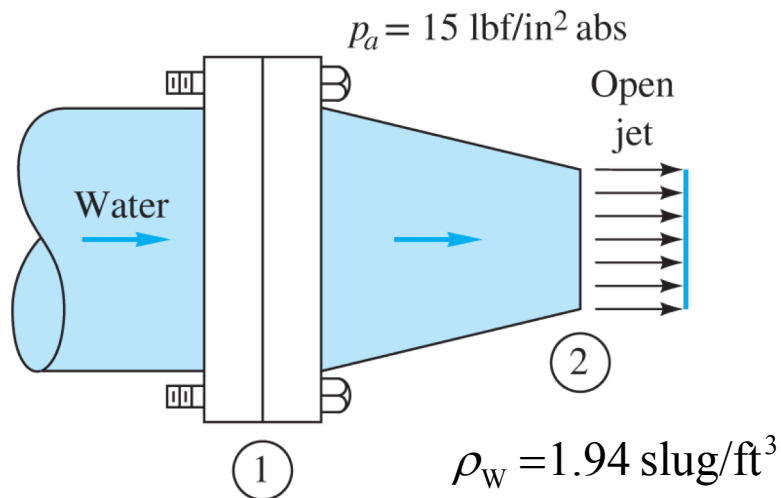


Anchoring Force to Hold an Elbow or Vane Stationary

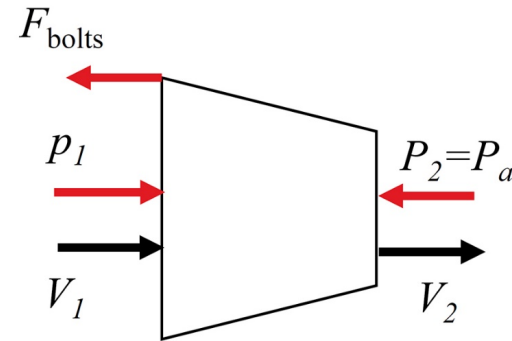
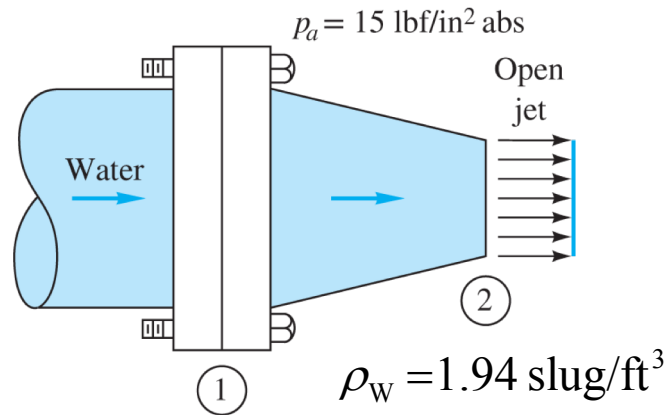
- When a liquid flow passes through an elbow that changes the direction of flow, or water jet deflected by a vane into different direction, it is crucial to determine the anchoring forces required to hold the elbow and vane stationary.

Example: The horizontal nozzle has $D_1 = 12$ in and $D_2 = 6$ in, with inlet pressure $p_1 = 38$ lbf/in² absolute and $V_2 = 56$ ft/s.

For water at 20°C, determine the horizontal force provided by the flange bolts.



Anchoring Force to Hold an Elbow or Vane Stationary



Solution: Select the outside of the nozzle plus the cut sections (1) and (2) as CV. Applying the mass balance eq.

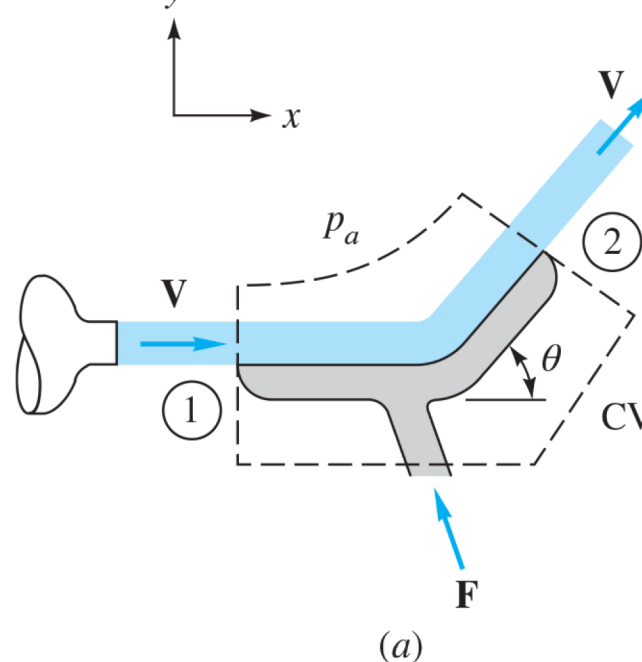
Applying the linear momentum eq.,

$\Rightarrow F_{\text{bolts}} = 1700 \text{ lbf}$

Anchoring Force to Hold an Elbow or Vane Stationary

Example: A fixed vane turns a water jet of area A through an angle θ without changing its velocity magnitude V . The flow is steady, pressure is p_a everywhere. (a) Find the components F_x and F_y of the applied vane force. (b) Express the force F and the angle ϕ between F and the horizontal in term of θ .

(Also, ignore the weight of fluid and the vane weight within the CV.)



Anchoring Force to Hold an Elbow or Vane Stationary

Solution:

(a) Select a CV cuts through the inlet and exit of the jet through the vane support --- vane friction is self-canceling.

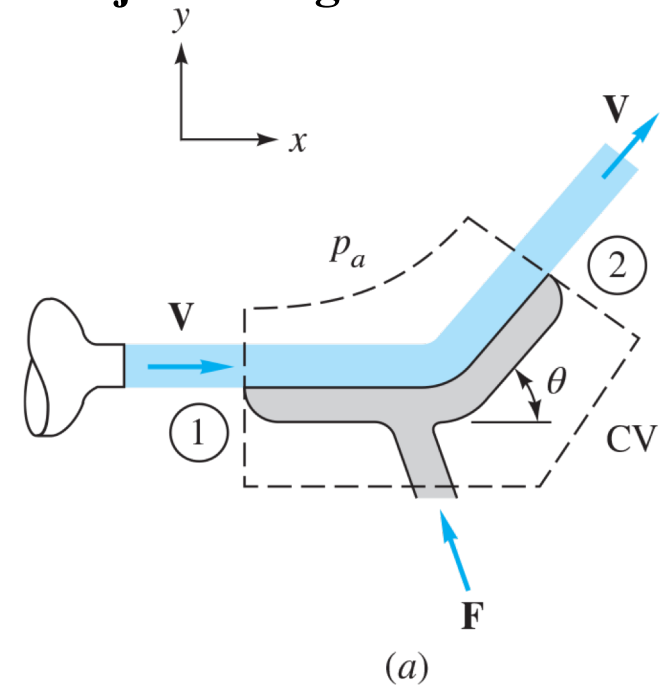
By the mass conservation,

Applying the momentum eq. and neglecting weight of fluid and vane

$$\mathbf{F} = \text{[]}$$

Components of the force in x and y direction are:

$$F_x = \text{[]} \quad F_y = \text{[]} \quad \text{where } \dot{m}V = \rho AV^2$$



Anchoring Force to Hold an Elbow or Vane Stationary

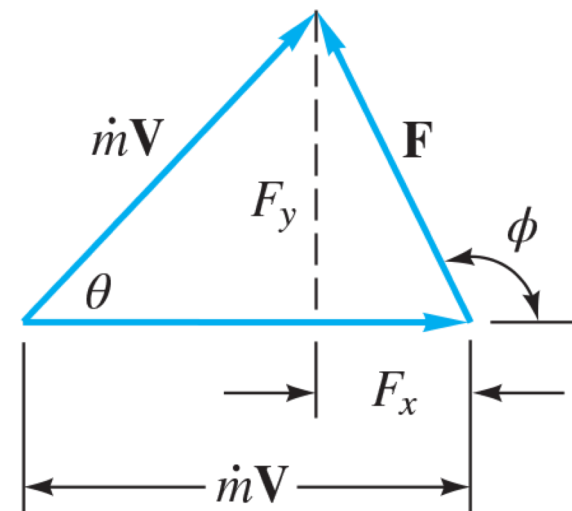
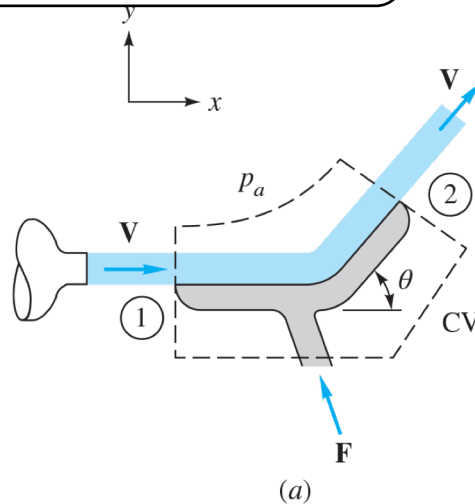
Solution (cont.):

(b) The resultant force magnitude is

$$F = \sqrt{F_x^2 + F_y^2} =$$

From the geometry,

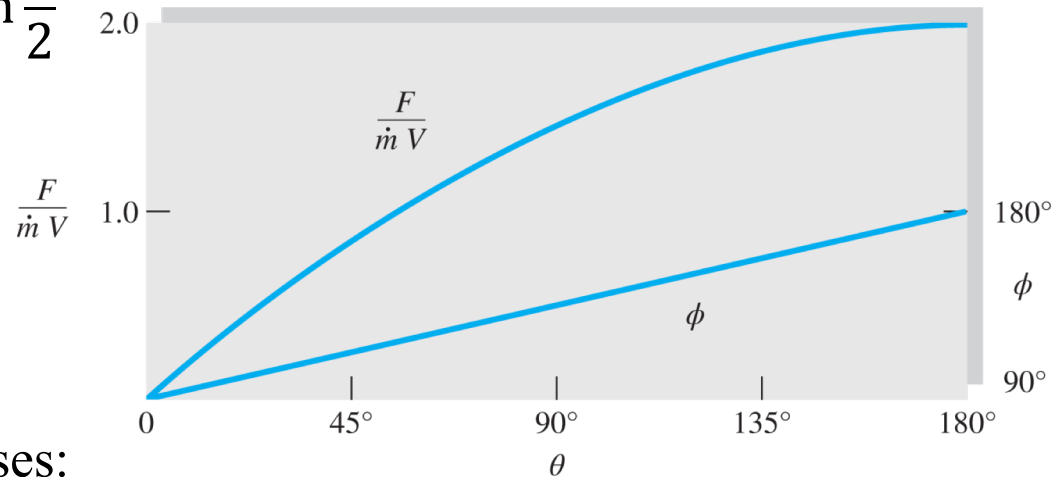
$$\phi =$$



Anchoring Force to Hold an Elbow or Vane Stationary

- Relation between F and θ

$$\frac{F}{\dot{m}V} = 2 \sin \frac{\theta}{2}$$



Two special cases:

- At $\theta = 180^\circ$, maximum force occurs.
The jet is turned around and thrown back in the opposite direction with its momentum reversed.
- At very small θ , we have $F = \dot{m}V\theta$, and $\phi = 90^\circ$
The force is linearly proportional to the turning angle, and acts normal to the jet (lift force).

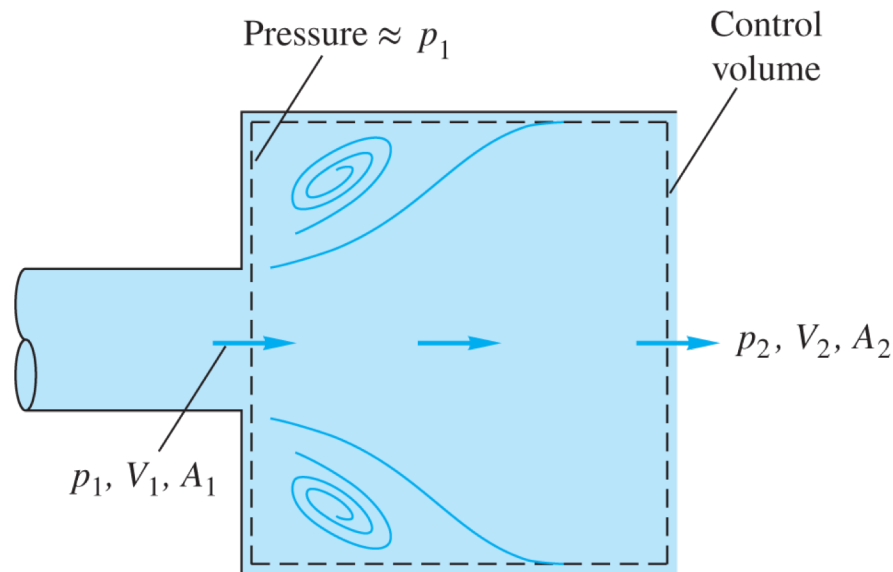
Friction Force Exerted by Solid Wall on Fluid Flow

- Many engineering applications need to find the friction force exerted by pipe wall on the fluid flow in a pipe or determine the drag force induced by a flat plate on the flow over it.

Example: A pipe flow suddenly expands from A_1 to A_2 .

Assume the flow is steady and incompressible, and $p_w \approx p_1$ on the corner annular ring as shown.

p_1, V_1, A_1, p_2, A_2 , and ρ are given, find V_2 and the wall friction.

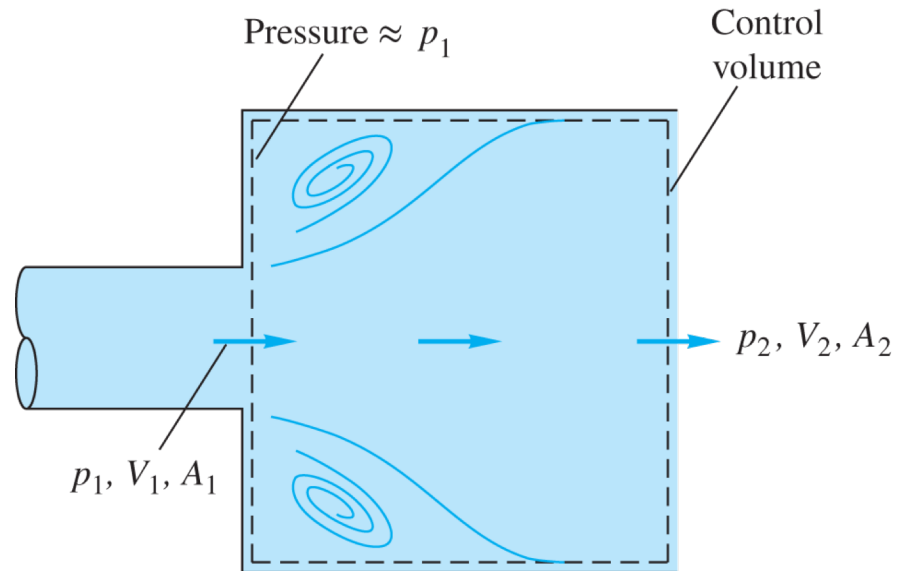


Friction Force Exerted by Solid Wall on Fluid Flow

Solution:

Select the inside of the expansion plus the cut sections (1) and (2) as CV

From the mass conservation,

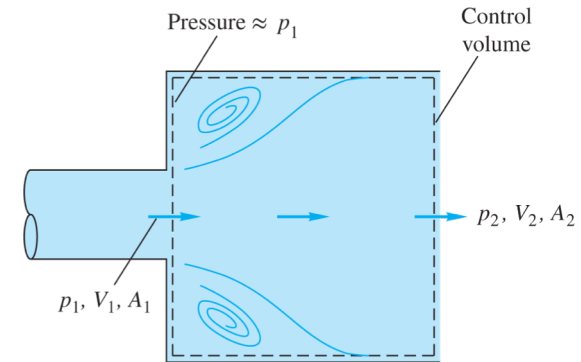


Friction Force Exerted by Solid Wall on Fluid Flow

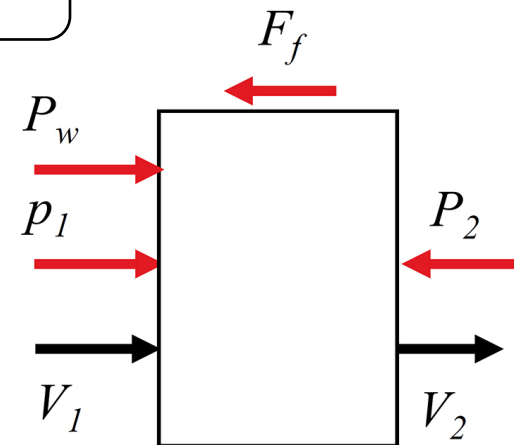
Solution (Con'd):

Applying the linear momentum eq.

$$\sum \mathbf{F} = \dot{m}(\mathbf{V}_2 - \mathbf{V}_1)$$



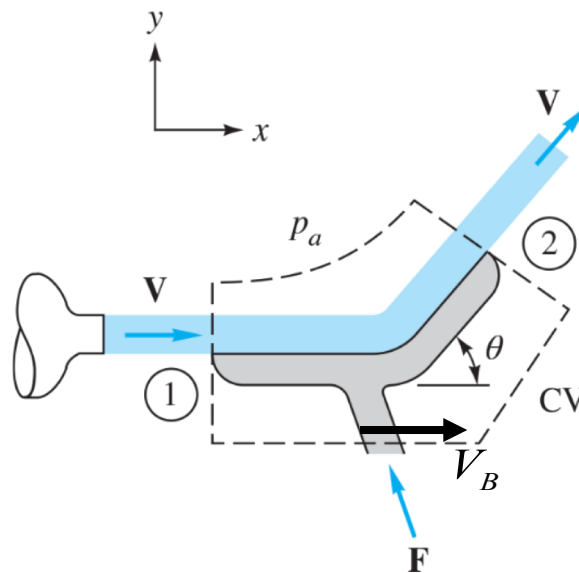
$$\Rightarrow F_f = A_2(p_1 - p_2) + \rho A_1 V_1^2 \left(1 - \frac{A_1}{A_2}\right)$$



Moving Control Volume

- A non-deforming control volume that translates at constant speed is inertial because there is no acceleration. In this case, the linear momentum equation is still applicable, except that the velocity V should be treated as the relative velocity.

Example 1: Re-consider the previous example assuming the vane is moving at a constant velocity V_B , find the components F_x and F_y of the applied vane force.



Select a CV cuts through the inlet and exit of the jet through the vane support

→ $V_r = \boxed{}$ (V_I and V_B are in the same direction.)

$$\dot{m}_1 = \dot{m}_2 = \dot{m} \quad \text{where} \quad \dot{m} = \left[\right]$$
$$\sum \mathbf{F} = \dot{m} (\mathbf{V}_2 - \mathbf{V}_1)$$

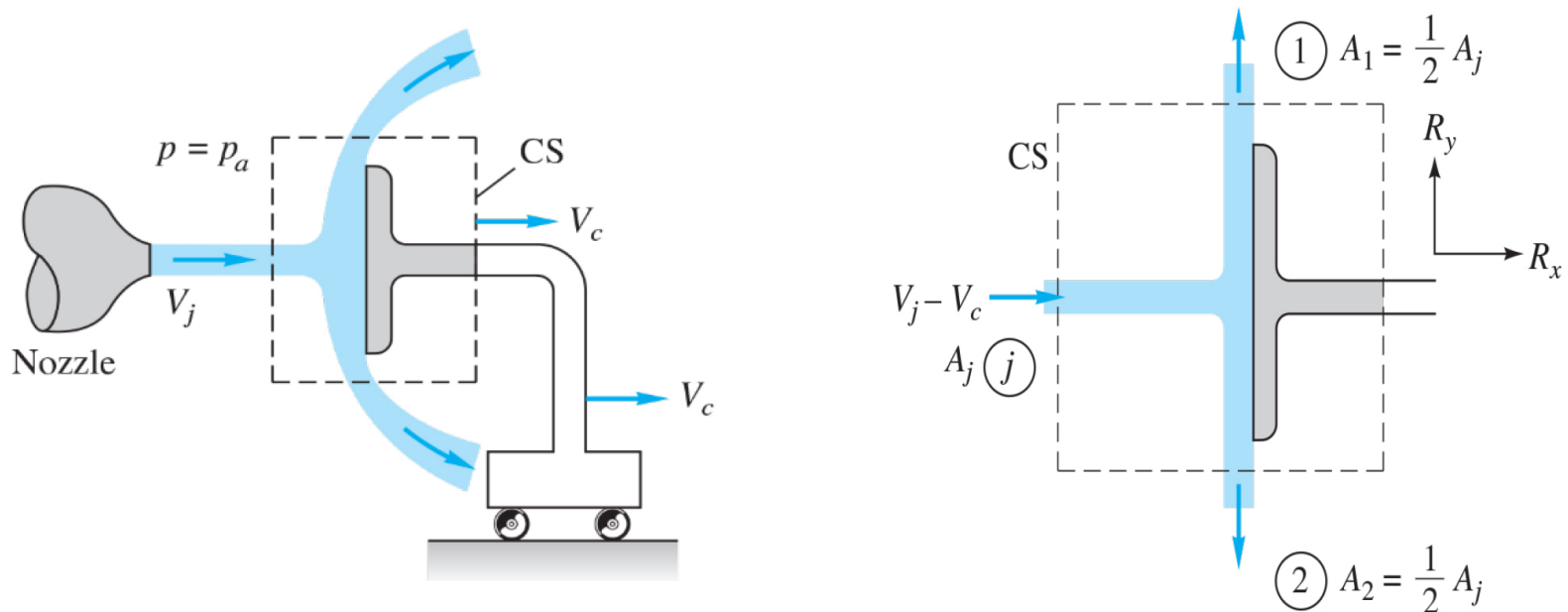
→ $F_x = \dot{m} V_r (\cos \theta - 1) = \dot{m} (V_1 - V_B)^2 A (\cos \theta - 1)$

$$F_v = \dot{m} V_r \sin \theta = \dot{m} (V_1 - V_B)^2 A \sin \theta$$



Moving Control Volume

Example: A water jet of velocity V_j impinges normal to a flat plate that moves at velocity V_c . Find the force required to keep the plate moving at constant velocity if $\rho_j = 1000 \text{ kg/m}^3$, $A_j = 3 \text{ cm}^2$, $V_j = 20 \text{ m/s}$, and $V_c = 15 \text{ m/s}$.



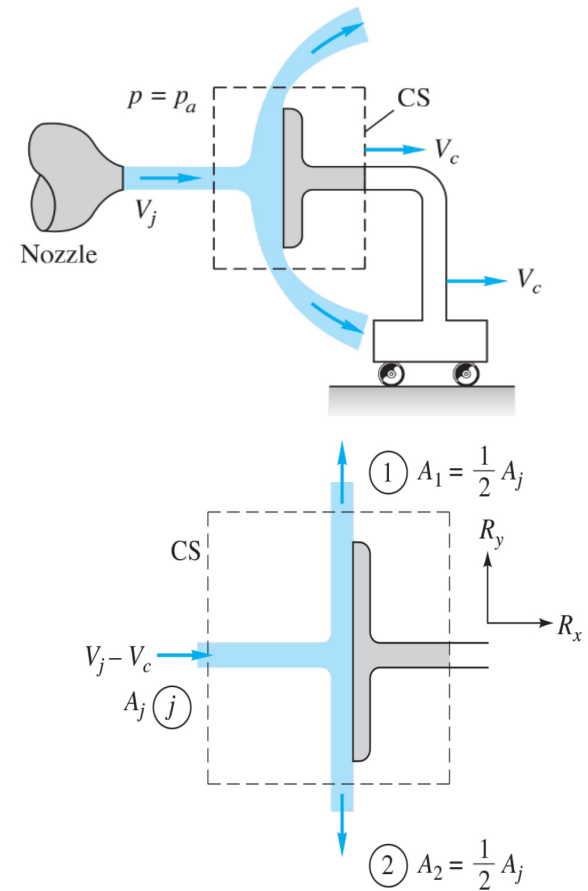
Moving Control Volume

Solution: Select CV cuts through the plate support.

Assumptions:

1. steady flow with respect to the moving plate,
2. negligible weight of jet and plate,
3. symmetric jet,
4. no lateral spreading.

By the mass conservation,



Moving Control Volume

Solution (continued):

Applying the momentum eq. $\mathbf{F} =$

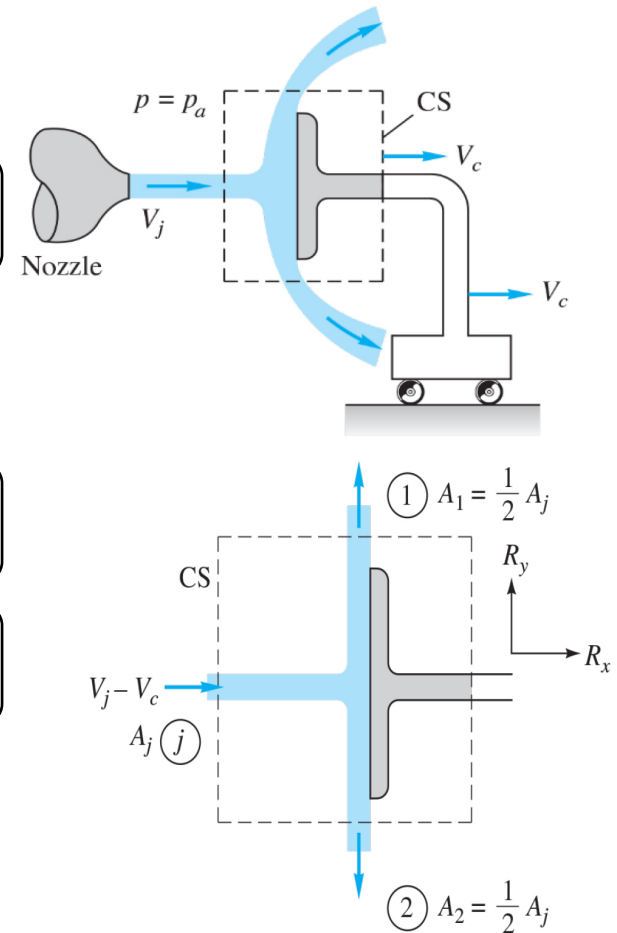
$$V_1 =$$

$$F_x =$$

$$\rightarrow R_x =$$

$$F_y = R_y =$$

$$\rightarrow$$



-
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Frictionless Flow: The Bernoulli Equation

- A classic linear momentum analysis between pressure, velocity, and elevation in a **frictionless** flow results in the **Bernoulli equation**.

$$\frac{p_1}{\rho} + \frac{1}{2} V_1^2 + gz_1 = \frac{p_2}{\rho} + \frac{1}{2} V_2^2 + gz_2 = \text{const} \quad (\text{steady form})$$

- The Bernoulli equation can be derived using the following restrictive assumptions:
 1. **Steady flow**: a common assumption for most flows.
 2. **Incompressible flow**: applicable for liquid flow, and gaseous flow with Mach number less than 0.3.
 3. **Frictionless flow**: restrictive---solid walls and mixing introduce friction.
 4. **Flow along a single streamline**: different streamlines may have different Bernoulli constants.

Frictionless Flow: The Bernoulli Equation

Conservation of mass equation

$$\frac{d}{dt} \left(\int_{CV} \rho dV \right) + \dot{m}_{out} - \dot{m}_{in} = 0 \approx \frac{\partial \rho}{\partial t} + d\dot{m}$$

$$d\dot{m} = d(\rho AV) = -\frac{\partial \rho}{\partial t} A ds$$

Linear momentum equation in the streamwise direction

$$\sum dF_s = \frac{d}{dt} \left(\int_{CV} \rho dV \right) + (\dot{m}V)_{out} - (\dot{m}V)_{in} \approx \frac{\partial}{\partial t} (\rho V) A ds + d(\dot{m}V)$$

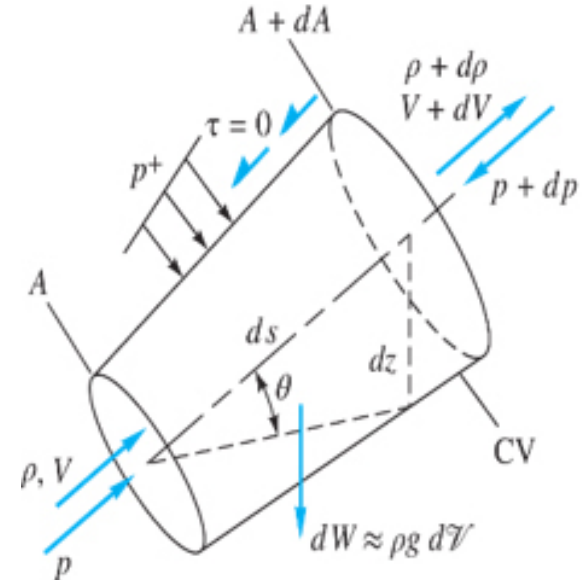
$V_s = V$ because s is the streamline direction

Neglecting shear force on the wall, only pressure and gravity forces remain.

$$dF_{s,grav} = -dW \sin \theta = -\gamma A ds \sin \theta = -\gamma A dz$$

$$dF_{s,press} = \frac{1}{2} dp dA - dp(A + dA) \approx -A dp$$

Elemental fixed streamtube CV



Frictionless Flow: The Bernoulli Equation

Linear momentum equation is rearranged as

$$\begin{aligned}\sum dF_s &= -\gamma A dz - A dp = \frac{\partial}{\partial t} (\rho V) A ds + d(\dot{m} V) \\ &= \cancel{\frac{\partial \rho}{\partial t} V A ds} + \frac{\partial V}{\partial t} \rho A ds + \dot{m} dV + \cancel{V d\dot{m}} \quad \text{(canceled by the continuity relation)}\end{aligned}$$

$$\frac{\partial V}{\partial t} ds + \frac{dp}{\rho} + V dV + g dz = 0, \left(\frac{\gamma}{\rho} = g \right)$$

Bernoulli's equation for unsteady frictionless flow along a streamline



Frictionless Flow: The Bernoulli Equation

For steady incompressible (constant-density) flow,

$$\frac{p_2 - p_1}{\rho} + \frac{1}{2}(V_2^2 - V_1^2) + g(z_2 - z_1) = 0$$

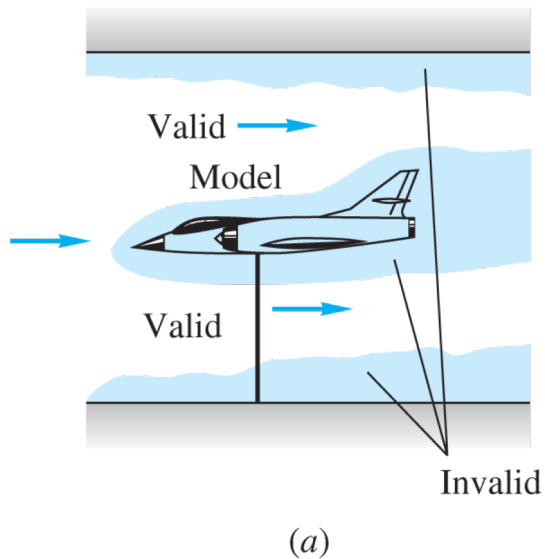
or



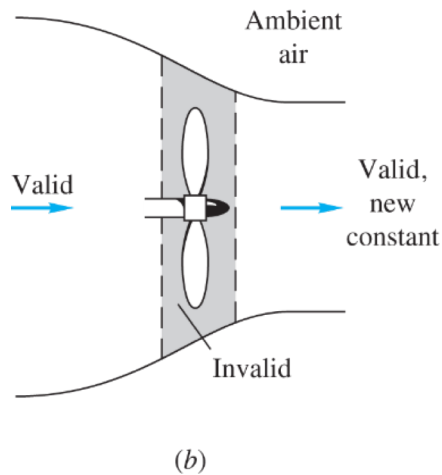
- For frictionless, and incompressible fluid, the Bernoulli equation may also be interpreted as an **idealized energy equation**. The changes from 1 to 2 in above equation represents *reversible* pressure work, kinetic and potential energy changes.

Restrictions on the Bernoulli Equation

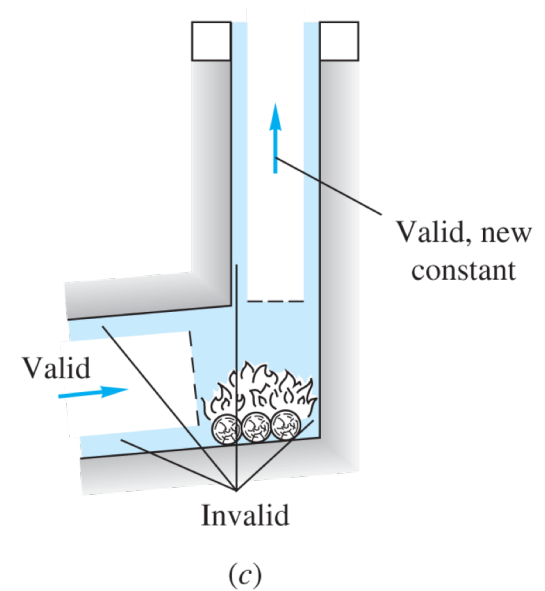
- Regions of validity and invalidity of the Bernoulli equation



Tunnel



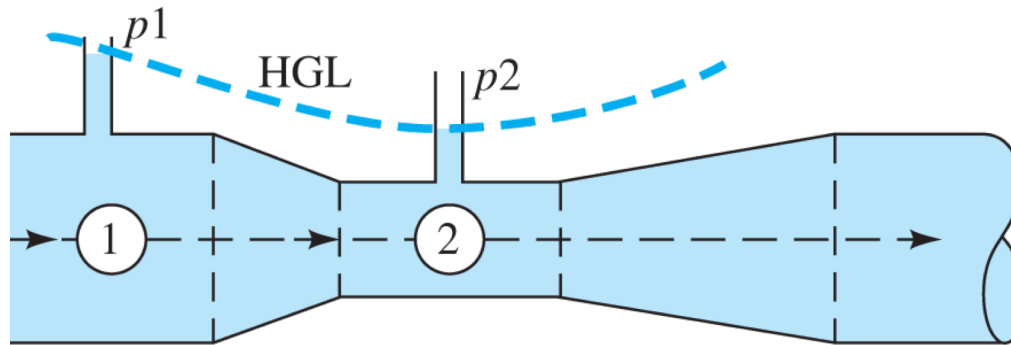
Propeller



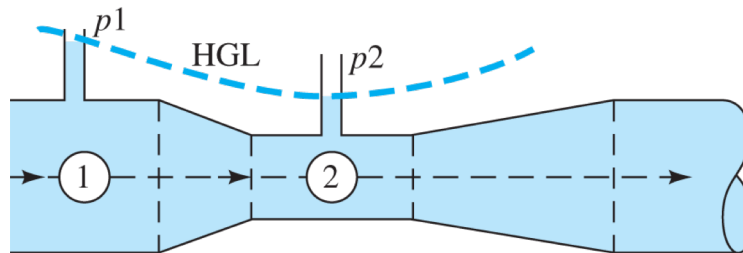
Chimney

Frictionless Flow: The Bernoulli Equation

Example: A constriction in a pipe will cause the velocity to rise and the pressure to fall at section 2 in the throat. The pressure difference is a measure of the flow rate through the pipe. The smoothly necked-down system is called a *venturi tube*. Find an expression for the mass flow in the tube as a function of the pressure change.



Frictionless Flow: The Bernoulli Equation



Solution: Apply Bernoulli eq. along the center streamline from section 1 to 2.

$$\frac{p_1}{\rho} + \frac{1}{2} V_1^2 + gz_1 = \frac{p_2}{\rho} + \frac{1}{2} V_2^2 + gz_2$$

If the tube is horizontal, $z_1 = z_2$ and solve for V_2 ,

From the mass conservation,

Combining the above equations, $V_2 = \sqrt{\frac{2\Delta p}{\rho(1-\beta^4)}}$

So the mass flow rate is $\dot{m} = \rho A_2 V_2 = \sqrt{\frac{2\rho\Delta p}{(1-\beta^4)}}$

The Angular Momentum Equation

- **The angular momentum:** $\mathbf{B} = \mathbf{H}_0$, the instantaneous angular momentum about O by integration over the mass dm is:

$$\mathbf{H}_0 = \int_{\text{syst}} (\mathbf{r} \times \mathbf{V}) dm \quad \text{so} \quad \beta = \frac{d\mathbf{H}_0}{dm} = \mathbf{r} \times \mathbf{V}$$

- For a nondeformable inertial control volume,

$$\frac{d(\mathbf{H}_0)_{\text{syst}}}{dt} = \sum \mathbf{M}_0 = \frac{d}{dt} \left(\int_{\text{CV}} (\mathbf{r} \times \mathbf{V}) \rho dV \right) + \int_{\text{CS}} (\mathbf{r} \times \mathbf{V}) \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

$$\text{or} \quad \sum \mathbf{M}_0 = \frac{d}{dt} \left(\int_{\text{CV}} (\mathbf{r} \times \mathbf{V}) \rho dV \right) + \int_{\text{CS}} (\mathbf{r} \times \mathbf{V}) \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

- Steady flow with 1-D inlets and outlets

$$\sum \mathbf{M}_0 = \sum_i (\mathbf{r}_i \times \mathbf{V}_i)_{\text{out}} \dot{m}_{i,\text{out}} - \sum_i (\mathbf{r}_i \times \mathbf{V}_i)_{\text{in}} \dot{m}_{i,\text{in}}$$

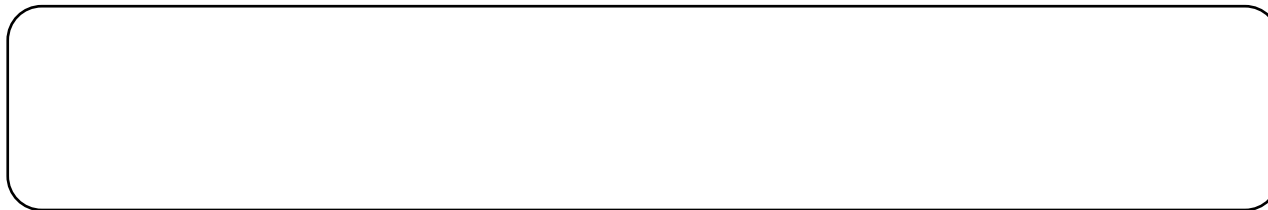
Angular momentum flow

The Energy Equation

- **Conservation of energy** is based on the 1st law of thermodynamics
- Conservation of energy: $B = E$, $\beta = dE/dm = e$, fixed CV

$$\left(\frac{dB_{\text{sys}}}{dt} = \frac{d}{dt} \left(\int_{\text{CV}} \beta \rho dV \right) + \int_{\text{CS}} \beta \rho (\mathbf{V} \cdot \mathbf{n}) dA \right)$$

(Net flux of heat into the CV) - (Rate of work done by the CV)
= (Rate of change of energy inside the CV)



Recall the sign convention: positive Q denotes heat added to the system
and positive W denotes work done by the system.

The Energy Equation

- The total system energy per unit mass e is expressed by

$$e = \underset{\text{Microscopic}}{e_{\text{internal}}} + \underset{\text{Macroscopic}}{e_{\text{kinetic}}} + e_{\text{potential}} = u + \frac{1}{2}V^2 + gz$$

- The work term is divided in three parts,

$$\dot{W} = \dot{W}_{\text{shaft}} + \dot{W}_{\text{press}} + \dot{W}_{\text{viscous}}$$

$$\text{with } \dot{W}_v \approx 0 \text{ and } \dot{W}_p = \int -p(-\mathbf{V} \cdot \mathbf{n})dA$$

$$\dot{W} = \dot{W}_s - \int p(-\mathbf{V} \cdot \mathbf{n})dA$$

(Shaft work is deliberately done by a machine, fan blade, piston, pump impeller through the control surface into the control volume)

- The heat transfer term is from conduction, radiation, and convection

The Energy Equation

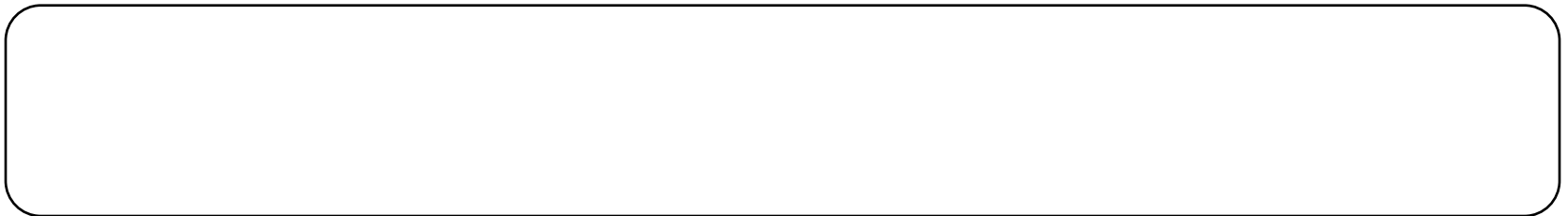
- Recall energy equation,

$$\frac{\delta Q}{\delta t} - \frac{\delta W}{\delta t} = \frac{dE_{\text{sys}}}{dt} = \frac{d}{dt} \left(\int_{\text{CV}} e \rho dV \right) + \int_{\text{CS}} e \rho (\mathbf{V} \cdot \mathbf{n}) dA \quad \text{and} \quad \dot{W} = \dot{W}_s - \int p (-\mathbf{V} \cdot \mathbf{n}) dA$$

- Integrate using Reynolds transform theorem,

$$\dot{Q} - \dot{W}_s = \frac{\partial}{\partial t} \left(\int_{\text{CV}} e \rho dV \right) + \int_{\text{CS}} \left(e + \frac{p}{\rho} \right) \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

- Energy equation for steady flow, 1-D inlets and outlets (1st law)



The Energy Equation

- One inlet/outlet (1-inlet, 2-outlet, $\dot{m}_1 = \dot{m}_2 = \dot{m}_3$),

$$\dot{Q} - \dot{W}_s = \sum \dot{m}_{\text{out}} \left(u + \frac{1}{2} V^2 + gz + \frac{p}{\rho} \right)_{\text{out}} - \sum \dot{m}_{\text{in}} \left(u + \frac{1}{2} V^2 + gz + \frac{p}{\rho} \right)_{\text{in}}$$

$$\rightarrow \dot{Q} - \dot{W}_s = \dot{m} \left[(h_2 - h_1) + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1) \right] \quad \text{where } h \equiv u + \frac{p}{\rho}$$

- Rearrange using the per unit mass form, (divide by $\dot{m}$)

$$q - w_s = h_2 - h_1 + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1) \quad \text{where } q = \frac{\dot{Q}}{\dot{m}} \quad w_s = \frac{\dot{W}_s}{\dot{m}}$$

- Head form of the energy equation, (*pressure head* = $\frac{p}{\gamma} = \frac{p}{\rho g}$)

$$h_q - h_s = \left(\frac{u_2}{g} + \frac{p_2}{\gamma} + \frac{V_2^2}{2g} + z_2 \right) - \left(\frac{u_1}{g} + \frac{p_1}{\gamma} + \frac{V_1^2}{2g} + z_1 \right) \quad \text{where } h_q = \frac{q}{g} \quad h_s = \frac{w_s}{g}$$

The Energy Equation

- Energy eq. for steady, incompressible flow with one inlet/outlet,

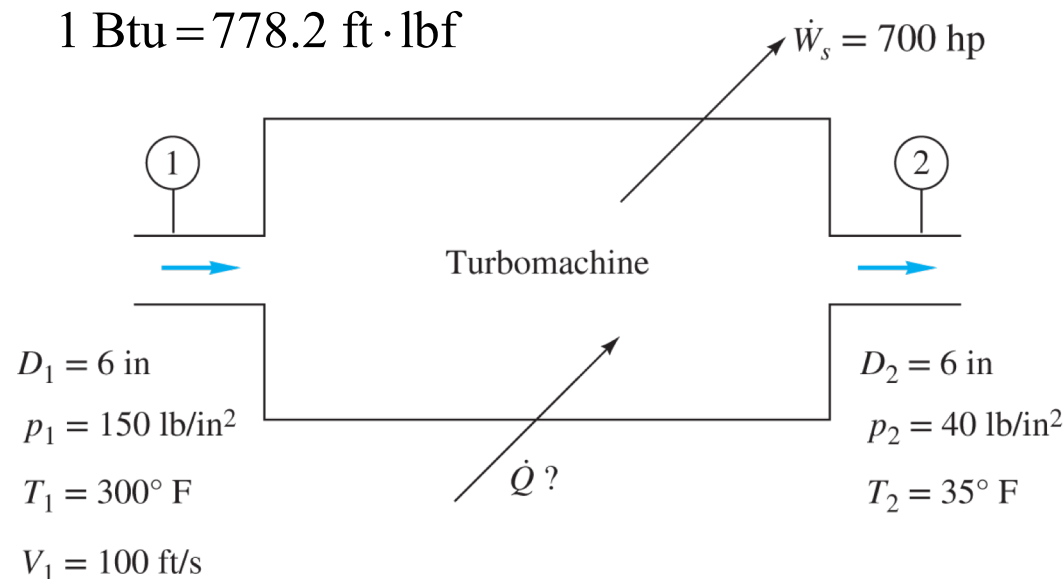
$$\left(\frac{p_1}{\gamma} + \frac{V_1^2}{2g} + z_1 \right)_{\text{in}} = \left(\frac{p_2}{\gamma} + \frac{V_2^2}{2g} + z_2 \right)_{\text{out}} + h_{\text{turbine}} - h_{\text{pump}} + h_f$$

Pressure head Velocity head Turbine head Pump head Head loss

where $h_f = \frac{(u_2 - u_1 - q)}{g}$

The Energy Equation

Example: Air ($R=1716, c_p=6003 \text{ ft} \cdot \text{lbf}/(\text{slug} \cdot ^\circ\text{R})$) flow steadily through a turbine that produce 700 hp of power. For the inlet and exit conditions shown, estimate (a) exit velocity V_2 and (b) the heat transfer rate in Btu/h.



The Energy Equation

Solution:

Select air in the turbine as **CV**, and assume air as ideal gas.

The inlet and exit densities,

$$\rho_1 = \boxed{} = \frac{150(144)}{1716(300 + 460)} = 0.0166 \text{ slug/ft}^3$$

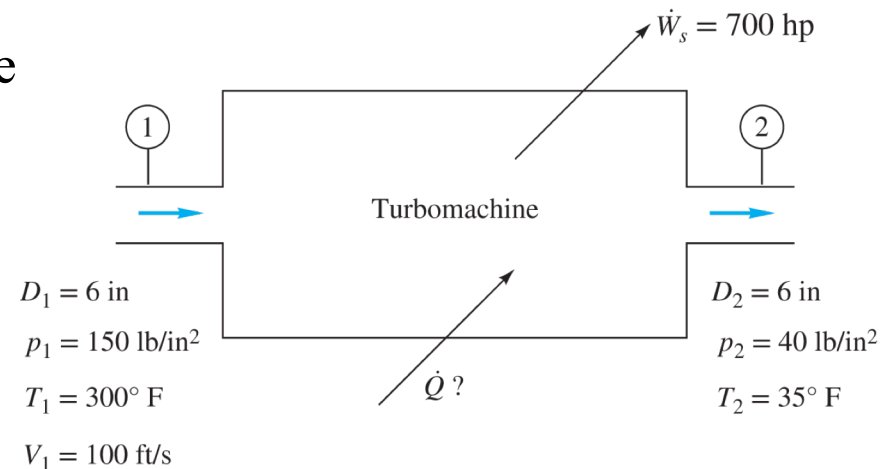
$$\rho_2 = \boxed{} = \frac{40(144)}{1716(35 + 460)} = 0.00679 \text{ slug/ft}^3$$

The mass flow rate is $\dot{m} = \boxed{} = (0.0166) \frac{\pi}{4} \left(\frac{6}{12} \right)^2 (100) = 0.325 \text{ slug/s}$

The exit velocity,

$$\dot{m} = 0.325 = \boxed{} = (0.00679) \frac{\pi}{4} \left(\frac{6}{12} \right)^2 V_2 \quad \rightarrow \quad V_2 = 244 \text{ ft/s}$$

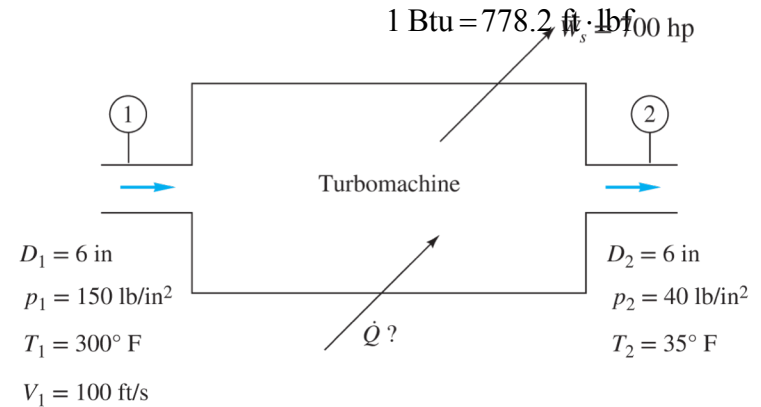
$$1 \text{ Btu} = 778.2 \text{ ft} \cdot \text{lbf}$$



The Energy Equation

$$\dot{m} = \rho_1 A_1 V_1 = (0.0166) \frac{\pi}{4} \left(\frac{6}{12} \right)^2 (100) = 0.325 \text{ slug/s}$$

$$V_2 = 244 \text{ ft/s}$$



$$\dot{Q} - \dot{W}_s = \sum \dot{m}_{\text{out}} \left(u + \frac{1}{2} V^2 + gz + \frac{p}{\rho} \right)_{\text{out}} - \sum \dot{m}_{\text{in}} \left(u + \frac{1}{2} V^2 + gz + \frac{p}{\rho} \right)_{\text{in}}$$

Steady flow energy equation applies with $z_1 = z_2$, $h = c_p T$,

$$\dot{Q} - \dot{W}_s = \dot{m} \left[c_p (T_2 - T_1) + \frac{V_2^2 - V_1^2}{2} \right]$$

$$\dot{Q} - 700(550) = 0.325 \left[6003(495 - 760) + \frac{244^2 - 100^2}{2} \right]$$

$$\dot{Q} = -125,000 \text{ ft} \cdot \text{lbf/s} = -578,000 \text{ Btu/h}$$

Summary

- All the laws of solid mechanics are written for a system, defined as an arbitrary quantity of mass of fixed identity.
- Reynolds transport theorem connects the mathematics between a system of individual masses and a region specified as the control volume (CV).

- The conservation of mass for steady and 1-D flow,

$$\sum (\rho_i A_i V_i)_{\text{out}} - \sum (\rho_i A_i V_i)_{\text{in}} = 0$$

- The linear momentum equation for steady and 1-D flow,

$$\sum \mathbf{F} = \sum_i (\dot{m}_i \mathbf{V}_i)_{\text{out}} - \sum_i (\dot{m}_i \mathbf{V}_i)_{\text{in}}$$

- Two main points for the CV analysis: 1. selection of a proper, clever, and workable CV; 2. reduction of the analysis that applies to the problem

- The energy eq. for steady, incompressible flow with one inlet/outlet

$$\left(\frac{p_1}{\gamma} + \frac{V_1^2}{2g} + z_1 \right)_{\text{in}} = \left(\frac{p_2}{\gamma} + \frac{V_2^2}{2g} + z_2 \right)_{\text{out}} + h_{\text{turbine}} - h_{\text{pump}} + h_f$$

- If the friction is negligible or tracing along a streamline in a fluid flow,

$$\frac{p_1}{\rho} + \frac{1}{2} V_1^2 + g z_1 = \frac{p_2}{\rho} + \frac{1}{2} V_2^2 + g z_2 = \text{const} \quad \text{Bernoulli equation}$$